7.7.3 Other tokenizers

As mentioned earlier, there are other tokenizers, and they are briefly described in table 7.4. You can find code examples with the book's files.

Table 7.4 Out-of-the-box tokenizers

Tokenizer	Description
`pattern`	Splits a field into tokens on a regex match. The default pattern is to split words when a non-word character is encountered.
`uax_url_email`	Parses fields and preserves URLs and emails. The URLs and emails are returned as they are, without any tokenization.
`whitespace`	Splits text into tokens when whitespace is encountered
`keyword`	Doesn't touch the tokens. This tokenizer returns the text as is.
`lowercase`	Splits text into tokens when a non-letter is encountered and lowercases the tokens
`path_hierarchy`	Splits hierarchical text such as filesystem folders into tokens based on path separators

The final component of an analyzer is a token filter. Its job is to work on the tokens spit out by the tokenizers. We briefly discuss the token filters in the next section.

7.8 Token filters

The tokens produced by tokenizers may need further enriching or enhancements such as lowercasing (or uppercasing) the tokens, providing synonyms, developing stemming words, removing apostrophes or other punctuation, and so on. Token filters work on tokens to perform such transformations.

Elasticsearch provides almost 50 token filters, and we can't discuss all of them here. We look at a few—feel free to reference the official documentation to learn about the rest of the token filters. We can test a token filter by simply attaching to a tokenizer and using it in the `_analyze` API call.

Listing 7.47 Adding a token filter with the tokenizer

```
GET _analyze
{
  "tokenizer" : "standard",
  "filter" : ["uppercase","reverse"],
  "text" : "bond"
}
```

The filter accepts an array of token filters; for example, here we provide the `uppercase` and `reverse` filters. The output is "DNOB" ("bond" is uppercased and reversed).

We can also attach filters to a custom analyzer, as the following listing shows. Then because we know how to attach token filters, we look at a few examples.